AI AND IOT BASED IRRIGATION SYSTEM USING SOIL MOISTURE

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***Abstract*— This paper proposes a smart irrigation system that utilizes Internet of Things (IoT) and Artificial Intelligence (AI) technologies to improve the efficiency and effectiveness of water usage in agriculture. The system incorporates IoT sensors to collect data on soil moisture and weather conditions, which is then analysed by AI algorithms to determine the optimal amount of water needed for the crops. The system can automatically control the irrigation system based on the values of the various environmental factors like humidity, temperature, light intensity and soil moisture which we can measure through sensors like the DHT-11 Humidity and Temperature sensor, Moisture sensor and LDR sensor, reducing water waste and increasing crop yield. The implementation of this smart irrigation system can lead to a more sustainable and efficient use of water resources in agriculture. We are using PLSR algorithm to train our model.**

Keywords— Internet of Things (IoT) and Artificial Intelligence (AI) technologies, temperature, humidity, soil moisture, light intensity, DHT-11 Temperature and humidity sensor, Moisture sensor, LDR sensor, PLSR Algorithm

I. INTRODUCTION

Agriculture is a vital sector that requires a significant amount of water resources for irrigation. With the increasing demand for food and the growing concern about water scarcity, it is crucial to optimize water usage in agriculture. The integration of Internet of Things (IoT) and Artificial Intelligence (AI) technologies can provide a solution to improve the efficiency and effectiveness of water usage in agriculture.

This paper proposes a smart irrigation system that uses IoT and AI technologies, specifically the Partial Least Squares Regression (PLSR) algorithm, to optimize water usage in agriculture. The system utilizes IoT sensors to collect data on soil moisture and weather conditions, which is then analysed by the PLSR algorithm to determine the optimal amount of water needed for the crops. The system can automatically control the irrigation system based on the results, reducing water waste and increasing crop yield.

The implementation of the smart irrigation system using the PLSR algorithm has the potential to revolutionize the way water is used in agriculture and can lead to a more sustainable and efficient use of water resources. This paper will discuss the design and implementation of the smart irrigation system and its potential impact on sustainable agriculture and water usage efficiency.

II. LITERATURE REVIEW

A literature review on smart irrigation using IoT and AI with PLSR algorithm would involve studying prior research and findings on the topics of IoT in agriculture, AI in irrigation, and PLSR as an algorithm for data analysis.

Research on IoT in agriculture has shown that the use of sensors and other IoT devices can help optimize irrigation systems by providing real-time data on soil moisture, weather conditions, and crop water needs. This can lead to improved water use efficiency, reduced water waste, and improved crop yields.

AI has been shown to be a valuable tool for decision-making in irrigation systems. By using machine learning algorithms such as PLSR, data from IoT devices can be analysed to make more accurate predictions about water needs and to optimize water use in real-time.

PLSR is a statistical technique for data analysis that has been applied in various domains, including agriculture. It can be used to analyse data from IoT devices and other sources to develop models for predicting water needs, optimizing irrigation schedules, and making other important decisions in smart irrigation systems.

Overall, the literature suggests that the use of IoT and AI in irrigation, along with the application of PLSR and other machine learning algorithms, has the potential to revolutionize the way we manage our water resources and grow our crops.

A. SURVEY OF EXISTING SYSTEM

Smart irrigation systems using IoT and AI have become a popular area of research due to their potential to improve the efficiency and effectiveness of irrigation. A literature survey of research in this field indicates that various techniques and technologies have been used to create smart irrigation systems, including machine learning algorithms, sensor networks, and cloud computing.

Several studies have proposed the use of IoT and AI to analyze data from sensors such as soil moisture sensors, temperature sensors, and humidity sensors to determine the water requirements of crops. Machine learning algorithms have been used to make predictions and decisions based on this data. In addition, cloud computing has been used to store and process data, making it accessible from anywhere.

Several hardware platforms have been used to implement smart irrigation systems, including Arduino, Raspberry Pi, and microcontrollers. Android applications have also been developed to enable remote control and monitoring of the irrigation systems.

Overall, smart irrigation systems using IoT and AI have the potential to significantly reduce water consumption while improving crop yield and quality. However, more research is needed to optimize the performance of these systems, particularly in terms of power consumption, data transmission, and security.

Here are a few existing smart irrigation systems using IoT and AI:

Netafim Smart Irrigation: An IoT-based system that uses weather data and soil moisture sensors to optimize watering schedules for crops.[1]

Smart Irrigation system using Internet of Things: An IoT-based smart irrigation system that uses sensors to collect data related to irrigation and stores it in cloud storage.[2]

Internet of Things Based Smart Irrigation: The system aims to provide a sustainable solution for irrigation by using IoT to automatically monitor and control the irrigation process using regression algorithm.[3]

Smart Irrigation Control System Using Wireless Sensor Network Via Internet-of-Things: Smart irrigation control system utilizes wireless sensor networks to collect data related to irrigation and then sends this data to a customized server. This server then serves as an Internet of Things (IoT) platform, which enables remote access to the data and allows for real-time monitoring and control of the irrigation process.[4]

Automated Irrigation and Crop Security System in Agriculture Using Internet of Things: The proposed system involves outfitting farmland with sensors, buzzers, and a scarecrow toy to collect data on various environmental factors, including temperature, humidity, soil moisture, and PIR (Passive Infrared Sensor). The collected data is then analyzed and monitored in real-time using Internet of Things (IoT) technology, with updates sent every 10 seconds.[5]

IoT Based Low Cost Smart Irrigation System: The Smart Irrigation System implements association with IoT using Smart wireless sensors to brings a great management system.[6]

IoT-Based Smart Irrigation Systems: We determine which parameters are monitored in the irrigation system in relation to water quantity and quality, soil properties and weather conditions. Provides an overview of the most commonly used nodes and radio technologies.

Smart farming: In this paper, they noted research work related to the concept of monitoring and advanced control for precision irrigation.

Internet of Things application for implementation of smart agriculture system: In this paper, the design and implement of a new wireless mobile robot based on the Internet of Things (IoT) to perform various tasks in the field.

Automated Irrigation System Using a Wireless Sensor Network and GPRS Module: Automatic irrigation systems have been developed to optimize crop water use.

These systems use different approaches, but all aim to conserve water, improve plant health, and reduce waste by optimizing irrigation schedules based on real-time data.

B. LIMITATIONS OF EXISTING SYSTEM

The limitations of smart irrigation systems using IoT and AI, and specifically the PLSR algorithm, could include:

1. High Cost: The cost of implementing a smart irrigation system with IoT and AI can be high due to the cost of sensors, hardware, and software.
2. Complexity: The integration of IoT and AI can be complex and may require specialized expertise to set up and maintain.
3. Data Accuracy: The accuracy of the data used to optimize irrigation schedules can impact the effectiveness of the system. This includes the accuracy of weather data, soil moisture sensors, and other sensors.
4. Algorithm Limitations: The PLSR algorithm may not be the most suitable algorithm for every application, and its accuracy may depend on the quality of the data used as input.
5. Limited User Control: Some systems may have limited options for user customization and control, reducing the flexibility of the system.
6. Internet Connectivity: Smart irrigation systems that rely on internet connectivity may be impacted by network outages or slow connectivity.

Overall, the limitations of smart irrigation systems using IoT and AI, including the PLSR algorithm, may impact the effectiveness and efficiency of the system.

III. PROPOSED SYSTEM

A. Framework

A framework for smart irrigation using IoT and AI using the PLSR (Partial Least Squares Regression) algorithm could include the following steps:

1. Data Collection: Collect data on soil moisture, temperature, humidity, and other relevant factors using IoT sensors. Data Pre-processing: Clean and process the collected data to remove any noise or outliers.
2. Model Development: Use the PLSR algorithm to develop a predictive model that takes into account the relationship between the input factors and the desired output (irrigation).
3. Model Validation: Validate the model using a separate dataset to ensure that it accurately predicts the irrigation needs.
4. Deployment: Deploy the model on an IoT platform, to control the irrigation system in real-time.
5. Monitoring: Continuously monitor the system and update the model as needed to ensure optimal irrigation and conserve water.
6. User interface: A user-friendly interface can be provided to the farmers to easily operate the system and receive alerts and suggestions.

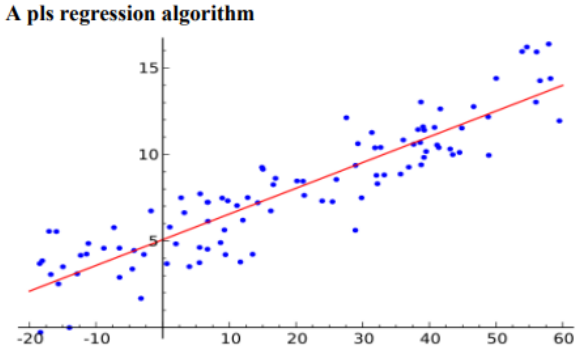
B. Algorithms

PLSR Algorithm

The goal of pls regression is to predict Y from X and explain their common structure. If Y is a vector and

X is full rank; this goal can be achieved using ordinary multiple regression. If the number of predictors is large compared to the number of observations, X is likely singular and the regression approach is no longer viable (that is, due to multicollinearity). Several approaches have been developed to address this problem. One approach is to eliminate some predictors (for example, using stepwise methods). Another approach, called principal component regression, is to do principal component analysis (pca) of the X matrix, then take her principal components of X and use the regressed variables. Y. The orthogonality of the principal components eliminates the multicollinearity problem. However, the problem of choosing the best subset of predictors remains. One possible strategy is to keep only some of the first components. But they were chosen to explain X, not Y, so there is nothing to guarantee that the principal components that "explain" X are related to Y. Specifically, pls regression looks for a set of components

(called latent vectors) that perform a joint decomposition of X and Y. These components are subject to constraint to explain as much of the covariance between X and Y as possible.



FORMULA USED

With one independent variable, we may write the regression equation as:

$$Y = a + bX + e$$

Where Y is an observed score on the dependent variable, a is the intercept, b is the slope, X is the observed score on the independent variable, and e is an error or residual.

We can extend this to any number of independent variables:

$$Y = a+b_1X_1+b_2X_2+\ldots. +b_kX_k+e \quad (3.1)$$

Note that we have k independent variables and a slope for each. We still have one error and one intercept. Again, we want to choose the estimates of a and b so as to minimize the sum of squared errors of prediction. The prediction equation is:

$$Y' = a+b_1X_1+b_2X_2+\ldots. +b_kX_k \quad (3.2)$$

Finding the values of b (the slopes) is tricky for k>2 independent variables, and you really need matrix algebra to see the computations. It's simpler for k=2 IVs, which we will discuss here. But the basic ideas are the same no matter how many independent variables you have. If you understand the meaning of the slopes with two independent variables, you will likely be good no matter how many you have.

For the one variable case, the calculation of b and a was:

$$b = \frac{\sum xy}{\sum x^2}$$

$$a = \bar{Y} - b\bar{X}$$

For the two-variable case:

$$b_1 = \frac{(\sum x_1^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - \left((\sum x_1 x_2)\right)^2}$$

And

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - \left((\sum x_1 x_2)\right)^2}$$

At this point, you should notice that all the terms from the one variable case appear in the two-variable case. In the two-variable case, the other X variable also appears in the equation. For example, X2 appears in the equation for b1. Note that terms corresponding to the variance of both X variables occur in the slopes. Also note that a term corresponding to the covariance of X1 and X2 (sum of deviation cross-products) also appears in the formula for the slope.

The equation for a with two independent variables is:

$$a = \bar{Y} - b_1\overline{X_1} - b_2\overline{X_2}$$

This equation is a straight-forward generalization of the case for one independent variable.

IV. METHODOLOGY

The system designed consists briefly of two major components:

1. Implementation of system using Internet of Things.
2. Artificial Intelligence to predict the amount of rainfall and the soil moisture content.

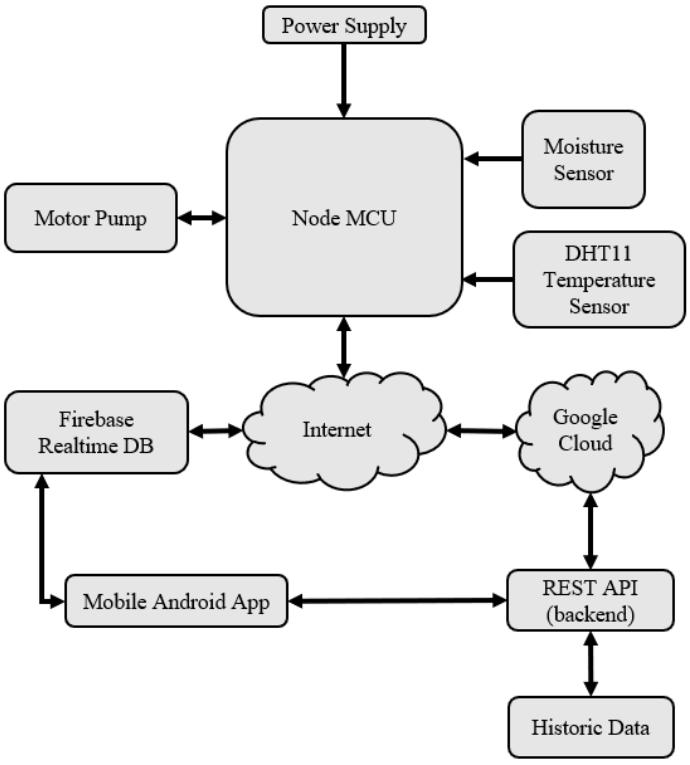


Fig. 1 Implementation of system using Internet of Things

A central system is used to control the nodes and determine when pumps connected to a particular node are turned on, in this implementation a Cloud API controller is used. The controller performs all functions using Python3 and JAVA scripts. The support script is imported into the main script, and the control logic calls methods in the hardening script as needed. Functions include communication with nodes, data preprocessing, and inference using the PLSR algorithm. Training is performed on a high-performance computer and then the trained model is deployed to the generating environment.

Each node consists of an ESP8266[8] predictive microcontroller board (NodeMCU) and an array of analog sensors. These sensors are tuned to receive weather information that is converted from the to a digital 10-bit integer using an A/D converter located on each NodeMCU. The algorithm converts this data into 32-bit floating point values required by the NodeMCU's own inference model. Each node includes supporting hardware to provide the necessary power to the for the NodeMCU and sensors, and to facilitate debugging in the event of a node failure. The power circuit consists of a 9V battery input connected to 3. Regulator 3V. The 3.3V output is then used to power the NodeMCU and sensors [13]. The normal Vcc line and GND line are used to power the for the components mentioned above. Specifically, 9V power is directly supplied to the

L293D motor driver IC to drive the water pump based on the input received from the NodeMCU. Implementations may also include one or more solar panels that power the NodeMCU and sensor using battery power only when the pump is needed. The preprocessed data is sent to the central controller using a GET HTTP request with additional bytes used to identify each individual node.

The nodes connect to the central controller via a common WiFi access point using the IPv4 protocol. Communication between the nodes and the central controller uses a client-server architecture. Nodes do not interact with each other. This implementation uses a cloud server on the central controller. The server expands the URL into a webhook. The node connects to the server via this webhook and sends data. An HTTP GET request is used to send the sensor data to the server after post-computation. By default server replication sends two messages to the requesting node. 1. Whether pumps connected to a particular node should be turned on and 2. How long the pumps should be turned on. The period is calculated using the output of the regression algorithm and each node sends a request to the server at 30 minute intervals. So a network is essentially a wireless LAN. Because each node operates independently of the others, an intermittent failure of a node will not crash the entire system. It also makes isolating faulty nodes easier. The central controller's IPv4 address is hardcoded into each NodeMCU. The central controller is assigned a static IPv4 address using the access point settings. Port used for communication is set to 8080 in this implementation.

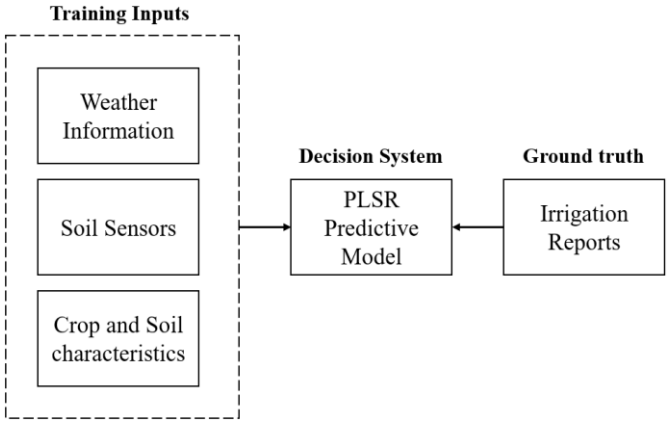


Fig. 2 Artificial Intelligence to predict the amount of rainfall and the soil moisture content

The figure 2 shows a schematic representation of a machine learning system. When trained on hard data, machine learning predictive decision systems can provide accurate estimates of a crop's irrigation requirements, given reliable and essential time information, including crop and soil condition data. Table 2 shows a set of possible factors that should be considered as inputs to the decision model. Evapotranspiration comes from the words "evaporation" and "transpiration". It refers to the process by which water is released into the atmosphere from soils and crops.

Scrutinizing the evapotranspiration rates of a crop is essential to estimating its irrigation requisites. The water to be irrigated to the crop is the magnitude of water level (moisture content) required to compensate for the evapotranspiration moisture loss from the field. Hence, ascertaining the evapotranspiration of a crop field is a major deciding factor in estimating the irrigation requisites of the crop. The FAO Penman-Monteith formula can be used to calculate the reference crop evapotranspiration (ET0) on a daily basis, using information from weather stations or alternatively weather sensors [20]. The

crop evapotranspiration can be calculated under standard conditions using the single crop coefficient formula given below:

$$ETc = K. ET0$$

Here Kc denotes the crop coefficient. It depends on various factors such as the crop type, crop evaporation, and crop growth stage and weather information.

We used the partial least squares regression (PLSR) algorithm [21] to develop the machine learning decision model. It is a statistical algorithm that identifies fundamental relationships between input variables and output variables. The predictor (input) variables marked with an X are defined as visually verifiable variables that are quantified to provide input to the decision model.

The replicated variable marked Y represents the outputs for which given inputs must be evaluated.

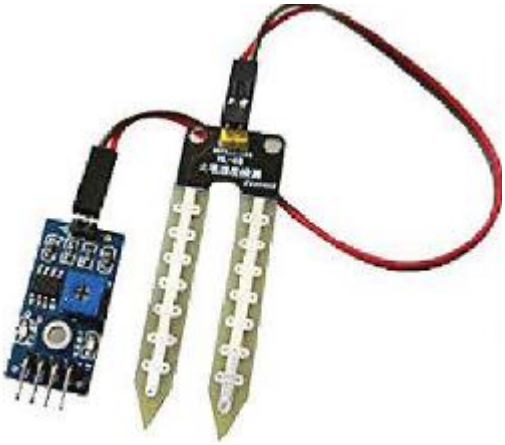


Fig. 3 Soil Moisture Sensor

Figure 3 shows the types of soil moisture sensors. This includes two tests by means of which an electric current penetrates the dirt, at which point the soil is inspected for clogging and the moisture level is checked. We know that being close to water makes it easier for dirt to conduct power effortlessly. This means that such types of soils have lower R (resistance) and dry soils have lower strength conductivity, i.e., dry soils hold more insurance than wet soils. The sensor is built on this strength characteristic. Need a point to convert resistance to voltage. This is done in a circuit inside the sensor that converts the resistance to a voltage.



Fig. 4 NodeMCU ESP8266

Figure 4 shows NodeMCU ESP8266 is a low-cost, open-source, IoT platform based on the ESP8266 Wi-Fi microcontroller. It integrates a microcontroller with Wi-Fi capabilities, making it easy to connect and control devices over the internet. It has on-board flash memory and a built-in voltage regulator, making it a compact and low-power solution for IoT projects. NodeMCU supports several programming languages and provides a development kit for users to quickly prototype and test their projects.



Fig. 5 DHT11 Sensor

Figure 5 shows the digital temperature and humidity sensor DHT11 is a composite sensor that contains a calibrated digital signal output of temperature and humidity. The technology of a dedicated digital modules collection and the temperature and humidity sensing technology are applied to ensure that the product has high reliability and excellent long-term stability. The sensor includes a resistive sense of wet component and an NTC temperature measurement device, and is connected with a high-performance 8-bit microcontroller



Fig. 6 Channel Relay Module

Figure 6 shows a relay is an electrically operated device. It has a control system and (also called input circuit or input contactor) and controlled system (also called output circuit or output cont. actor). It is frequently used in automatic control circuit. To put it simply, it is an automatic switch to controlling a high-current circuit with a low-current signal.

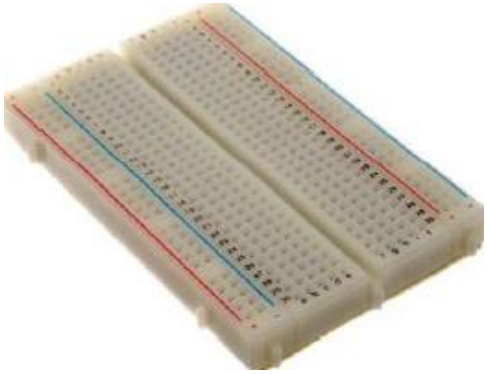


Fig. 7 Breadboard



Fig. 8 Jumper Wire

Figure 7 shows a breadboard is a more solid invention of a preliminary template with a test circuit design. A bounce wire (commonly called a jumper (figure 8), jumper wire, jumper interface, Dupont wire, or Dupont interface, named after one of the creators) is a wire or those corporate connections in his connection. or any end commonly used to connect a breadboard or other model or section of a test circuit to other hardware or components without soldering (or without them, as the case may be) paste (essentially "tinned"). Individual bob wires are connected by embedding the "end connector" into the original path, providing them with a breadboard, printed circuit board header connector, or touch test fixture.

V. IMPLEMENTATION

The NodeMCU board is used to collect the data from the soil moisture sensor and the DHT i.e Temperature and Humidity. The collected data from the sensors and different weather API is then processed by AI algorithms to determine the optimal amount of water required for the plants. The relay module is used to control the water pump that supplies the water to the plants. The entire system is connected to the internet, which allows it to be monitored and controlled from anywhere using an Android App.

STEPS:

Circuit diagram and connections are designed as shown in fig 9. Step by step process is described below:

- Connect the soil moisture sensor, DHT11 sensor to the NodeMCU board using jumper wires.
- Connect the relay module to the NodeMCU board using jumper wires.
- Connect the water pump to the relay module using jumper wires.
- Install the Arduino IDE and the NodeMCU 8266 board driver on your computer.
- Open the Arduino IDE and create a new project.
- In the Arduino IDE, go to Sketch -> Include Library -> Manage Libraries. Search for the Firebase Arduino library and install it.
- Upload the code to the NodeMCU board.
- Create a Firebase project and a new Firebase Realtime Database.
- Install Android Studio and Install the latest or above version 10 O.S
- Open the IDE and create a new project
- Connect the Andrioid app with the Firebase Project so it can receive the data
- Make Custom input for different plants and crops.

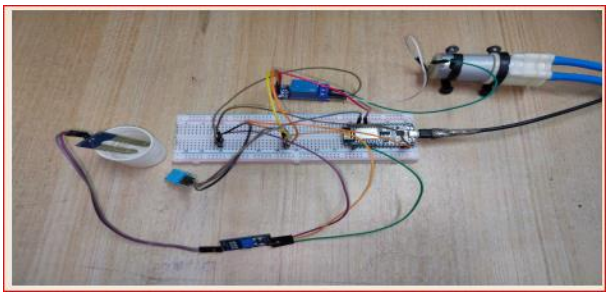


Fig.9 Circuit diagram for proposed model

VI. RESULTS

The water pup can be switched on and switched off according to the soil moisture levels which automate the process of irrigation as shown in fig.10.The system uses information from the soil moisture sensor to irrigate the soil which helps to prevent over irrigation or under irrigation of soil to avoid crop damage. The farm owners can monitor the process online through mobile application as shown in fig.11.



Fig.10 Proposed working model design

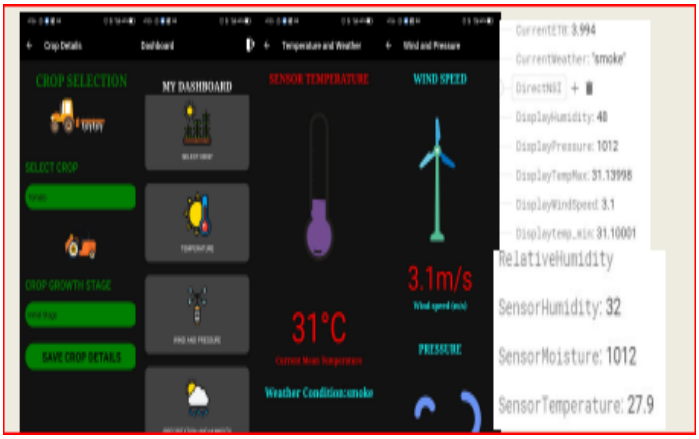


Fig.10 Monitoring using Android application

VII. CONCLUSION

Intelligent irrigation systems with IoT technology are highly effective solutions for reducing water waste and maximizing yields. By integrating IoT sensors, real-time weather data and soil moisture measurements, the system can optimize irrigation schedules and provide accurate and efficient irrigation. This results in lower water bills, improved plant health, and less strain on local water resources. In addition, the remote monitoring and control capabilities provided by IoT-powered smart irrigation systems make it more convenient and user-friendly for farmers. Overall, using IoT technology to implement intelligent irrigation systems is a step towards more sustainable and productive agriculture.

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